

GENRA: Enhancing Zero-shot Retrieval with Rank Aggregation

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IMAGE NOT PROVIDED (Image 1)

Figure 1: [Image 1 placeholder from source]

Abstract

Large Language Models (LLMs) have been shown to effectively perform zero-shot document retrieval, a process that typically consists of two steps: i) retrieving relevant documents, and ii) re-ranking them based on their relevance to the query. This paper presents GENRA, a new approach to zero-shot document retrieval that incorporates rank aggregation to improve retrieval effectiveness. Given a query, GENRA first utilizes LLMs to generate informative passages that capture the query’s intent. These passages are then employed to guide the retrieval process, selecting similar documents from the corpus. Next, we use LLMs again for a second refinement step. This step can be configured for either direct relevance assessment of each retrieved document or for re-ranking the retrieved documents. Ultimately, both approaches ensure that only the most relevant documents are kept. Upon this filtered set of documents, we perform multi-document retrieval, generating individual rankings for each document. As a final step, GENRA leverages rank aggregation, combining the individual rankings to produce a single refined ranking. Extensive experiments on benchmark datasets demonstrate that GENRA improves existing approaches, highlighting the effect...

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